

Hospital Operations & Revenue Analytics

Professional Portfolio Project Report

This portfolio-ready report summarizes an end-to-end hospital analytics project using Excel for cleaning, PostgreSQL for querying, and Power BI for visualization. The project analyzes patient flow, appointments, billing, treatments, and revenue to produce actionable business insights.

Project Workflow

Dataset → Excel Cleaning → PostgreSQL → SQL Analysis → Power BI Dashboard → Business Insights

Tools

Excel, PostgreSQL, SQL, Power BI

Dataset

Patients, Doctors, Appointments, Treatments and Billing tables.

Data Cleaning

Removed duplicates, handled missing values, standardized formats and categorical values, validated dates.

SQL Skills Demonstrated

Joins, Aggregations, GROUP BY, Subqueries, Window Functions, Ranking, Monthly Trends, Revenue Analysis.

Dashboard KPIs

Total Appointments: 200

Total Patients: 50

Revenue: \$551K

Average Billing: \$2.76K

Key Insights

- March recorded the highest appointment volume.
- Revenue is concentrated in a few treatments.
- Workload among doctors is uneven.
- Failed/pending payments delay cash flow.

Business Recommendations

Improve payment processing, balance doctor workload, diversify treatment offerings, investigate low-performing months.

Skills Demonstrated

Data Cleaning, SQL, Data Modeling, Power BI, KPI Design, Business Analysis, Dashboard Development.

Future Enhancements

Revenue forecasting, patient demand prediction, real-time dashboard, departmental analytics.